

CA-PFL: Client-adaptive Parameter-efficient Fine-tuning for Personalized Federated Learning

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Abstract

Large Language Models (LLMs) show excellent capabilities in complex language understanding and generation tasks, but their deployment on edge devices faces inherent contradictions between model scale and resource constraints, data privacy and performance optimization. Personalized Federated Learning (PFL) offers a promising solution by addressing gradient misalignment and communication overhead in traditional federated learning. Yet, existing PFL methods for large-scale models often rely on static architectures or manually defined configurations, limiting flexibility and adaptability. To this end, we propose CA-PFL, a client-adaptive parameter-efficient fine-tuning approach that dynamically generates personalization strategies through variational sparse control: 1) dynamically assigning LoRA rank based on the local data features distribution to establish sparsity aligned with the quantitative relationship of heterogeneous data; 2) utilizing variational Bayesian prior to estimate the optimal rank configuration, compatible with federated aggregation; and 3) reducing the communication cost through random Fourier feature (RFF) projection. Experiments show that CA-PFL achieves an average accuracy of 77.77% and 79.89% on the fine-tuned Llama-3.2B and Mistral-7B models, respectively, with a communication cost of only 22.4% of the baseline method, while maintaining the lowest client performance variance, realizing the Pareto optimization of privacy and performance.

CCS Concepts

• **Computing methodologies** → **Distributed computing methodologies.**

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Keywords

Personalized Federated Learning; Parameter-Efficient Fine-Tuning; LoRA; Large Language Models; Communication Efficiency

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1 Introduction

The rapid advancement of large language models (LLMs, e.g., Llama [63], GPT-4 [51]) has enabled breakthroughs in web-based intelligent services and applications, facilitating scenarios such as personalized content recommendation [38], conversational search systems [8], and adaptive web interfaces [67]. However, the deployment of these models in web-scale federated environments, where user data resides on heterogeneous client devices with stringent privacy constraints, gives rise to significant challenges. Traditional centralized web services require sharing sensitive user interactions and browsing data, violating privacy regulations such as GDPR [2], while direct on-device fine-tuning is often infeasible due to the massive computational demands of modern LLMs and the resource limitations of web clients. Addressing these challenges is crucial for enabling privacy-preserving, efficient, and personalized large model services across web applications and platforms.

Recent advances in Parameter-Efficient Fine-Tuning (PEFT [23]) techniques, particularly Low-Rank Adaptation (LoRA [24]), have emerged as promising solutions by freezing pre-training weights and updating only a few low-rank matrices. When used in conjunction with Personalized Federated Learning (PFL [61]), these techniques can theoretically enable privacy-preserving model personalization, while federated learning further enhances privacy by aggregating model updates instead of raw user data [74]. However, existing approaches struggle to balance personalization and efficiency in heterogeneous web environments [20]. The most notable problems stem from rigid sparsity control and static personalized strategies that are not capable of accommodating the dynamic properties of heterogeneous user behaviors across web clients and the extreme diversity of device capabilities ranging from mobile

browsers to desktop applications communicating with servers in a federated framework.

The first challenge in deploying a large model personalized fine-tuning framework for web-scale LLM applications is *the adaptive personalization for varying data distributions among heterogeneous web clients*. Existing PFL works on large models fine-tuning strategies typically employ uniform sparsity strategies [57], assigning fixed LoRA rank across all clients regardless of their task characteristics or user interaction patterns. This uniform strategy leads to either over-sparsification for complex web tasks (e.g., conversational agents, personalized recommendation) or under-sparsification for simple ones (e.g., form auto-completion, basic classification), failing to adapt to diverse user behaviors and evolving web application requirements. This limitation undermines their applicability in dynamic web scenarios. While manual layer-wise configurations [5] offer partial relief, they require extensive per-application tuning and still struggle to keep pace with shifting user interests and interaction patterns. Similarly, statically controlled personalization strategies [9] are ill-suited for the dynamic nature of web applications and heterogeneous client environments, with a heavy reliance on expert experience. Traditional adaptive personalization strategies [15] cannot be directly transferred to web-deployed large model scenarios as they cannot well adapt to the high-dimensional parameter space and non-linear feature interactions in large language foundation models [48]. As a result, existing PFL methods provide insufficient personalization for complex web-scale LLM tasks, which urgently necessitate *a federated framework for dynamically generated personalized adaptive sparsity strategies tailored to web environments*.

Another fundamental challenge is *achieving communication efficiency without compromising personalized performance in web-scale deployments*. Existing PEFT-based federated learning schemes still face significant communication bottlenecks [30] in practical web deployments, although they reduce the transmission scale through low-rank decoupling or parameter aggregation. For large models with enormous parameters (e.g., ViT-L/16 [13] and RoBERTa-large [42] coupled models with more than 300 million parameters), the traditional PFL full-parameter fine-tuning approaches [4] involve transmitting the complete model gradient, and the one-time communication overhead may exceed 1.2 GB, which is substantially above the bandwidth capacity of mobile web clients, especially those on cellular networks. While existing sparsity strategies reduce the parameter scale, adopting a static compression ratio [11] leads to more rounds to converge under non-independent and identically distributed (Non-IID) data. A typical scenario in web applications involves users with distinct interests and behaviors, which in turn increases the overall communication costs. The LoRA-based personalized federated approach lacks the consideration of communication-computation co-optimization and cannot dynamically adjust the communication strategy [65] according to the data distribution, network conditions (4G vs. 5G vs. WiFi), and device resources (mobile vs. desktop browsers), leading to inefficient communication in resource-constrained web scenarios. Therefore, there is an imminent demand to *construct an efficient solution that can simultaneously balance communication overhead and model accuracy for web-scale federated learning*, especially under the non-convex optimization scenarios in federated learning settings.

To overcome these challenges, we propose client-adaptive parameter-efficient fine-tuning for personalized federated learning (CA-PFL), a novel framework that incorporates variational Bayesian LoRA rank estimation with Fourier feature-driven personalization, specifically designed for web-scale LLM deployment. We construct a *variational Bayesian LoRA aggregation mechanism* to address the first challenge, which interprets local updates as probabilistic priors, completing the initialization of a personalized model based on the client data distribution and user interaction patterns. The variational prior model accomplishes LLM-related rank distributions among web clients, thus enabling adaptation for multi-client complex data and diverse web applications. This approach not only enhances convergence with theoretical guarantees for local variational rank estimation, but also effectively balances the privacy-utility tradeoffs critical for web services under privacy regulations. For the second challenge, we introduce a *variational sparsity controller* that maps the client data distribution to a hierarchical LoRA rank by layer via a stochastic Fourier projection. In contrast to predefined sparsity ratios, each web client autonomously determines the optimal layer sparsity by regularizing the latent vectors to prevent excessive sparsification of key parameters, dynamically adapts the sparsity strategies to accommodate network conditions and device capabilities, and ensures efficient resource utilization without sacrificing accuracy. Unlike directly transmitting the LoRA rank, CA-PFL only transmits the compact controller parameters, which significantly reduces the communication scale while realizing better privacy protection—particularly beneficial for mobile web clients with limited bandwidth. Our main contributions are summarized as follows:

- We present a web-scale federated PEFT framework that establishes dynamic coupling between sparsity and personalization, addressing the unique challenges of heterogeneous web user populations.
- We introduce a variational Bayesian-based federated LoRA optimizer with theoretical guarantees for local variational rank estimation, explicitly optimizing the sparsity adaptation under non-IID data distributions.
- We conduct extensive experiments across diverse base models and common-sense reasoning datasets, demonstrating Pareto-optimal performance in the communication-accuracy tradeoff critical for web applications.
- To facilitate reproducibility and future research, we have released our implementation code at <https://github.com/YanDang/CA-PFL>.

2 Related Works

2.1 Large Language Models (LLMs)

Large Language Models (LLMs [1, 33, 54]) are renowned for their powerful capabilities in natural language understanding and generation tasks. Their core architectures primarily consist of transformer-based architectures and embedding layers, connectors [33] (which map language-model features into the language model space), and large language models [6, 63] that achieve contextual alignment and reasoning through multi-stage pre-training and alignment fine-tuning. Represented by CLIP [54], which maximizes the similarity

of matched image-text pairs and minimizes the similarity of mismatched pairs through contrastive loss, it has become a benchmark approach for language modeling alignment. Flamingo [1] designs interleaved vision-text attention modules that dynamically adjust the interaction weights between visual tokens and text tokens, significantly improving the coherence of long-context text generation tasks. Despite demonstrating impressive capabilities in LLM-based tasks, the performance of these models is significantly compromised in real deployment environments due to resource constraints on edge devices [37] and distributional differences in massive data [59].

2.2 Parameter-efficient Fine-tuning

In response to the challenges of high training costs and difficult edge deployment for large models, PEFT techniques [23] have emerged as a research hotspot. PEFT achieves task adaptation by freezing the main parameters of the pretrained model and optimizing only a small number of new modules. Existing work can be roughly categorized into three types [12]: fine-tuning methods through the introduction of an adapter (Serial Adapter [52]) or soft-tuning (Prefix-tuning [36]), which do not modify the model itself but instead introduce an extra module as a medium for fine-tuning; selective parameter fine-tuning, represented by SAM [75], which does not require additional parameters but selects only a small subset of the parameters to be fine-tuned during downstream task training; and re-parameterized fine-tuning [44], which refers to the conversion of model parameters between two equivalent forms. LoRA [24] introduces additional low-rank trainable parameters during training, which are subsequently combined with the original model for inference. Among these PEFT methods, LoRA and its derivative algorithms [58] have garnered significant attention for their structural simplicity, theoretical validity, and broad compatibility with foundational models [6]. BayesTune [31] encourages the emergence of small-scale parameters by setting a Laplace prior for each coefficient. LoRETTA [68] dynamically adjusts the rank allocation strategy of the tensor decomposition through tensor-column decompositions to significantly reduce the parameter computational complexity. Overall, while these methods adjust the rank assignment of low-rank matrices to improve the parameter fine-tuning efficiency and model performance, the lower bound of generalization for the dynamic rank assignment task [34] has yet to be rigorously proven. Moreover, parameter updates in the low-rank space restrict the higher-order nonlinear interactions [71] between features that are often involved in complex language modeling tasks.

2.3 Personalized Federated Learning

In conventional Federated Learning (FL) frameworks [45], local data across clients frequently manifest Non-IID [73] characteristics. With the migration of user behavior patterns, the global model on the server faces dual challenges of generalization and dynamic adaptation [56], necessitating robust frameworks that balance generalization, robustness, and fairness [17, 26, 69]. Personalized Federated Learning (PFL [61]) is a solution to the issue of insufficient global model adaptability caused by client data heterogeneity by constructing personalized models locally on the clients [28, 62]. Existing research can be categorized based on technical approaches as

follows: Customized model design for clients based on parameter decoupling [3] or hybrid models [25, 37], Meta-learning-driven [16] personalized optimization methods, Knowledge distillation [39] or teacher-student model-based transfer methods, Dynamic adaptation strategies [66]. The design of personalized models for the client is typically achieved by separating the functional layers of the model. FedPer [3] freezes the shared feature extractor in the model and fine-tunes the classifier to achieve local model personalization. LG-FedAvg [37] sets the model head as a personalized module, trained locally on the client, while the model backbone is shared through server aggregation.

Although this manual configuration of personalized layers can enhance adaptability to heterogeneous data, the shared layers may suffer from feature shifts due to data heterogeneity [35], lacking flexibility and generalization. Meta-learning [18] is a "Learn-to-Learn" approach aimed at finding an optimal parameter strategy that can adapt to global tasks. FedMeta [7] optimizes local client training strategies by adjusting learning rate schedules through meta-learning. Although meta-learning enables efficient personalization through rapid task adaptation, it incurs high computational costs and relies heavily on centralized coordination by the server. Knowledge distillation [22] and teacher-student models share a similar concept, involving both the distillation and the lightweight transformation of complex models to transfer knowledge to client models. FedGKT [21] designs a lightweight local student model that interacts with a complex teacher model through distillation loss constraints for knowledge transfer. In addition, some approaches employ dynamic adaptation strategies for parameter-level personalization. DynaFed [53] dynamically adjusts the proportion of trainable parameters based on local data entropy, allocating more trainable parameters to clients with higher data entropy.

Recent studies have integrated LoRA rank adaptation with federated learning, using PEFT methods for client personalization. FedEx-LoRA [57] incorporates a residual error term in frozen weights to correct deviations between parameter updates and ideal solutions, achieving precise aggregation. But for non-IID data, residual term based on an assumed linear model may not be applicable. FedIT [72] focuses on large-scale model instruction tuning by employing low-rank adapters on clients to fine-tune large language models. Static adapters, though, may result in an inability to adapt to dynamic client data distribution changes. LoRA-FAIR [5] introduces a correction term on the server to mitigate aggregation bias. In privacy scenarios, corrections based on global data statistics lack availability. pFedLoRA [70] deploys small local adapters per client to capture data heterogeneity, complemented by a global adapter on the server for knowledge exchange. Instead, fixed adapter parameters do not allow for dynamic adaptation to both simple and complex tasks. FFA-LoRA [32] exclusively updates and aggregates zero-initialized weights, reducing almost half communication costs while minimizing noise amplification in differential privacy. A zero-initialization, however, slows down the convergence of initial training and introduces additional communication overhead.

Although recent approaches like FLoRA [64] have explored heterogeneous low-rank adaptations, most work still retain static sparsity [40] control in personalization strategies, failing to establish dynamic mapping between sparse matrices and data distribution

drift. Furthermore, the theoretical foundation for low-rank parameter fine-tuning remains underdeveloped, and the unconstrained residual terms may lead to potential conflicts between local client training and global model aggregation objectives [41].

3 Methodology

3.1 Preliminaries and Motivation

Federated Aggregation with LoRA. In a traditional federated learning framework (FedAvg [46]), assume that the system contains K web clients, each client k holds the private dataset $\mathcal{D}_k$. The server maintains the global model parameters θ_g , the k -th client fine-tunes on the local data $\mathcal{D}_k$ to obtain the updated parameters θ_k , and the t -th round of parameter aggregation can be expressed as:

$$\theta_g^{t+1} = \sum_{k=1}^K \frac{|\mathcal{D}_k|}{|\mathcal{D}|} \theta_k^{(t)}, \quad |\mathcal{D}| = \sum_{k=1}^K |\mathcal{D}_k|, \quad (1)$$

where $|\mathcal{D}_k|$ is the amount of data for client k . Considering the scenario of collaboratively training a large language model using a federated system, each client performs local fine-tuning and then sends the updated full parameters back to the server for weighted aggregation. This paradigm faces severe communication bottlenecks and privacy risks when the parameter dimension d becomes extremely large. To address this challenge, we incorporate the Low-Rank Adaptation (LoRA [24]) mechanism, which constrains model updates to a low-dimensional subspace. Specifically, LoRA freezes the model raw parameters $\theta \in \mathbb{R}^{d \times e}$ without updating, while only training the low-rank adapter matrix $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times e}$ ($r \leq \min(d, e)$). Therefore, the actual parameter updates can be expressed as:

$$\theta' = \theta + \frac{\alpha_{lora}}{r} BA, \quad (2)$$

where α_{lora} is a hyperparameter. LoRA reduces communication pressure and privacy leakage risk, but still relies on static rank definitions, resulting in personalized models that fail to reach optimal solutions.

Bayesian Sparse Fine-tuning. According to the BayesTune [31] framework, variational Bayesian estimation employs probability distributions to model parameter uncertainty and updates the prior distribution with observed data to obtain the posterior distribution. For model parameters θ , it introduces a Laplace prior:

$$p(\theta|\kappa) = \prod_{i=1}^d p(\theta_i|\kappa_i) = \prod_{i=1}^d \mathcal{L}(\theta_i, \bar{\theta}_i, \kappa_i), \quad (3)$$

where $\mathcal{L}(x; \mu, b)$ is the Laplace distribution. The model parameters have their mean set to the pretrained values $\bar{\theta}$, while the scale parameter κ_i controls the degree of deviation from initial parameters. The significance of the scale parameter is to express an a priori preference for parameter deviation from the pretrained value, it determines the strength of the penalty for deviation of the parameter from the ideal. Based on empirical Bayesian inference [14], it is known that given the web client data $\mathcal{D}_i$, we can infer the value of κ_i that best fits the data distribution, and at the same time, we can achieve the sparsification of κ_i for the sparse matrix update of the global model. A Gamma hyper-prior is further imposed on κ_i to promote sparsity. Most κ_i approach zero, while a few larger

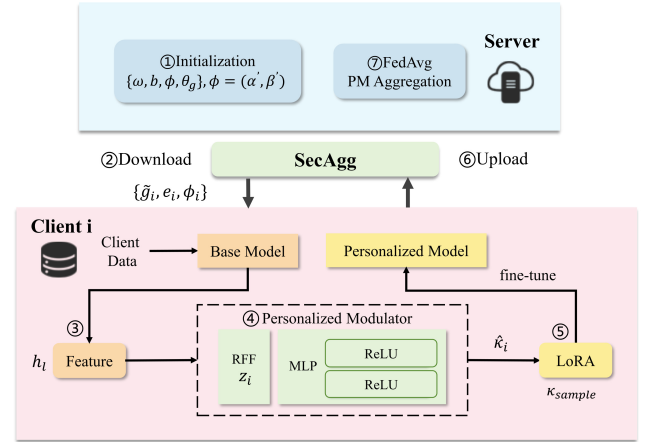


Figure 1: CA-PFL's process of generating personalized strategies in each round of training

κ_i identify crucial parameters requiring updates. The hyper-prior based on the gamma distribution can be expressed as:

$$p(\kappa|\alpha, \beta) = \prod_{i=1}^d p(\kappa_i|\alpha, \beta) = \prod_{i=1}^d \mathcal{G}(\kappa_i|\alpha, \beta), \quad (4)$$

where α is the shape parameter of the gamma distribution, β is the inverse scale parameter of the gamma distribution, and $\mathcal{G}(\cdot)$ denotes the probability density function of the gamma distribution. This framework provides a well-theorized sparsity auto-learning mechanism that avoids heuristic rules while maintaining computational efficiency.

3.2 Variational Bayesian Prior-Based Parameter Estimation for LoRA

In order to establish different LoRA rank selection strategies for different clients, we model the client personalized parameter fine-tuning problem as a LoRA sparse rank selection problem. Specifically, we decompose each LoRA rank matrix into three parts, finding a potential vector $\kappa_{l,i}$ between B and A matrices: $\Delta\theta_i = B\kappa_{l,i}A$. The potential vector is defined as the rank scale of the l -th module among the L LoRA modules.

In practical scenarios, we find that the weight gradient optimization is unstable due to the non-conductivity of the absolute value term of the Laplace prior. And the Laplace regular tends to produce strictly sparse solutions, which may excessively inhibit parameter updates and reduce the flexibility of the model in practical applications. Therefore, we use Huber loss [27] replacement, $L2$ approximation near the zero point and linear loss away from it, which is denoted as:

$$L_{smooth} = \sum_{l=1}^L \sum_{i=1}^r \begin{cases} \frac{\|B_{l,i}\|_1^2}{2\kappa_{l,i}} & \text{if } \|B_{l,i}\|_1 \leq \delta \\ \frac{\delta}{\kappa_{l,i}} \left(\|B_{l,i}\|_1 - \frac{\delta}{2} \right) & \text{otherwise,} \end{cases} \quad (5)$$

where δ is the Huber loss threshold that controls the switching boundary of the $L1/L2$ norm. Smooth $L1$ is more flexible than the Laplace prior to avoid over-sparsity when the task requires significant updates to some of the parameters, e.g., adapting a new task-critical

module. To support a multi-peaked distribution of the data, we optimize the Gamma hyper-prior as a variational distribution. Based on the variational KL scatter, the true posterior delay is approximated by the variational distribution $q(\kappa; \phi)$. The optimization objective can be expressed as:

$$KL(q(\kappa; \phi) || p(\kappa)) = \sum_{l=1}^L \sum_{i=1}^r \left[\frac{\alpha_g}{\alpha'} - \log \frac{\beta_g}{\beta'} - \psi(\alpha') + \log \frac{\Gamma(\alpha_g)}{\Gamma(\alpha')} + \alpha' \left(\frac{\beta'}{\beta_g} - 1 \right) \right], \quad (6)$$

where $q(\kappa; \phi)$ denotes the variational distribution of shape parameter α' , rate parameter β' , $p(\kappa)$ denotes the prior distribution of shape α_g , rate β_g , and the KL scatter $KL(q || p) = \mathbb{E}_q[\log q(\kappa) - \log p(\kappa)]$. $\psi(\cdot)$ is the digamma function, denoting the derivative of the logarithmic term of the gamma distribution. $\Gamma(\cdot)$ is the gamma function that satisfies $\Gamma(x) = (x-1)!$ with respect to a positive integer x . Variational inference is optimized by gradient descent and is more suitable for large-scale models than Markov Chain Monte Carlo (MCMC) [19]. Variational Parameters ϕ automatically learns the distribution of each LoRA scale parameter κ_i , eliminating the need to manually set hyper-parameters. When the parameters are extremely large (e.g., ViT [13], RoBERTa [42]), variational inference significantly reduces the computational overhead while maintaining the adaptive learning ability of sparsity. Combining the sparse regular term with the variational constraints, the objective function of the joint optimization can be expressed as:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{CE}(\theta; \mathcal{D}) + \alpha_s L_{smooth}(B, \kappa) + \alpha_p KL(p || q) + \alpha_f \|\theta - \theta_g\|_{2, \Omega}, \quad (7)$$

where $\alpha_s, \alpha_p, \alpha_f$ are the hyper-parameters, $\mathcal{L}_{CE}(\theta; \mathcal{D})$ denotes the task loss, which we use to guarantee the classification performance of the model on the dataset $\mathcal{D}$ with the cross-entropy loss. $L_{smooth}(B, \kappa)$ is the sparse regularization loss for parameter-level sparsification. $KL(p || q)$ denotes the variational constraint term for adaptively learning the sparsity strength κ , and $\|\theta - \theta_g\|_{2, \Omega}$ is the federated constraint term, which serves to constrain the difference between the local model and the global model in the federated learning. Ω is the significant parameter index for dynamic sparsification.

Our optimization framework achieves efficient training and deployment of models through multi-objective co-optimization. For distributed learning scenarios, the optional federated constraint term then supports collaborative fine-tuning across nodes by constraining the differences between local model parameters and the global model. All optimization objectives adopt differentiable design and support end-to-end training based on stochastic gradient descent or stochastic gradient Langevin dynamics, which not only ensures the numerical stability of the optimization process, but also flexibly adapts to the needs of different training scenarios. This integrated design allows the model to be both sparse, adaptive, and distributed collaborative while maintaining performance advantages.

3.3 Personalization Strategies for Dynamic Sparsification

In federated learning scenarios, using traditional fully-connected MLPs (3.2MB) for local personalized generation on web client devices presents two critical challenges: frequent model parameter transmissions incur high costs for bandwidth-constrained clients, and small local training datasets easily lead to parameter overfitting. To address these issues, we designed a solution based on Random Fourier Feature (RFF) projection matrices z_i :

$$z_i = \sqrt{\frac{2}{D}} [\cos(\omega^T h_i + b), \sin(\omega^T h_i + b)], \quad (8)$$

$$\hat{\kappa}_i = MLP(z_i),$$

where h_i represents the original features, ω is the random projection matrix updated through global aggregation on the server, b is the phase shift, and the scaling factor ensures the stability of the feature variance. A two-layer ReLU [49] network is employed for predictive structural transformation. RFF leverages Bochner's theorem [55] to map inputs into a fixed-dimensional Fourier feature space, approximating kernel function computations. By projecting the original high-dimensional features h_i into a fixed-dimensional space D , the subsequent MLP only needs to process low-dimensional features, significantly reducing the parameter count.

In terms of parameter scale, only the projection matrix ω and phase shift b need to be stored, cutting the parameter size by 75% (0.8MB). The periodic nature of cosine/sine functions inherently suppresses input noise, making them more robust than operations such as ReLU or pooling. Additionally, the built-in regularization term and high-frequency random projections naturally mitigate overfitting, akin to implicit data augmentation. This approach is computationally efficient for clients, requiring only matrix multiplication and trigonometric calculations, making it well-suited for edge device deployment. Figure 1 gives the running process of CA-PFL.

3.4 Theoretical Foundation and Error Analysis of Random Fourier Features

Random Fourier Features (RFF) provide a theoretically sound approach to approximate shift-invariant kernels via Bochner's theorem [55]. Given a shift-invariant kernel function $k(\mathbf{x}, \mathbf{y}) = k(\mathbf{x} - \mathbf{y})$, its Fourier transform can be expressed as the expectation over a spectral density $p(\omega)$:

$$k(\mathbf{x} - \mathbf{y}) = \mathbb{E}_{\omega \sim p(\omega)} \left[e^{i\omega^T (\mathbf{x} - \mathbf{y})} \right]. \quad (9)$$

By Monte Carlo sampling, RFF constructs an explicit low-dimensional mapping by drawing D random frequency vectors $\omega_j \sim p(\omega)$ and uniform phase shifts $b_j \sim \text{Uniform}(0, 2\pi)$:

$$\phi_{\text{RFF}}(\mathbf{x}) = \sqrt{\frac{2}{D}} \left[\cos(\omega_j^T \mathbf{x} + b_j) \right]_{j=1}^D, \quad (10)$$

such that the kernel function is approximated as:

$$k(\mathbf{x}, \mathbf{y}) \approx \phi_{\text{RFF}}(\mathbf{x})^T \phi_{\text{RFF}}(\mathbf{y}). \quad (11)$$

Theoretical bounds on approximation error can be derived from the properties of random projections. Let $\hat{k}(\mathbf{x}, \mathbf{y}) = \phi_{\text{RFF}}(\mathbf{x})^T \phi_{\text{RFF}}(\mathbf{y})$

denote the RFF-approximated kernel. The mean squared error (MSE) between $\hat{k}$ and the true kernel k satisfies:

$$\mathbb{E}[(\hat{k} - k)^2] \leq \frac{2}{D} \left(\mathbb{E}_{\omega} [\cos^2(\omega^\top(\mathbf{x} - \mathbf{y}))] - k^2 \right). \quad (12)$$

For the Gaussian kernel $k(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\sigma^2}\right)$, where the spectral density $p(\omega)$ follows $\mathcal{N}(\mathbf{0}, \sigma^{-2}\mathbf{I})$, the error bound is simplifies as:

$$\mathbb{E}[(\hat{k} - k)^2] \leq \frac{2}{D} \left(1 - \exp\left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{\sigma^2}\right) \right). \quad (13)$$

This demonstrates that the approximation error decreases at a rate of $O(1/\sqrt{D})$ as the RFF dimension D increases. In distributed learning, clients compute the empirical kernel mean embedding (EKME) of their local data:

$$\mu_{\text{local}} = \frac{1}{n} \sum_{i=1}^n \phi_{\text{RFF}}(\mathbf{x}_i), \quad (14)$$

which implicitly encodes first- and second-order statistics of the data distribution in a low-dimensional space while preserving privacy (since random projections prevent inversion back to raw data). After aggregating local embeddings μ_{local} from clients, the global distributional distance metric (e.g., maximum mean discrepancy, MMD) remains consistent:

$$\text{MMD}(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}} \approx \|\mu_{P,\text{RFF}} - \mu_{Q,\text{RFF}}\|_2. \quad (15)$$

In summary, RFF provides a computationally efficient and theoretically rigorous method for representing data distributions, with controllable approximation error and strong compatibility with the communication efficiency and privacy requirements of distributed learning.

3.5 Computational Overhead Analysis

A critical aspect of any federated learning framework designed for mobile edge devices is ensuring that local computations remain manageable. We provide a detailed complexity analysis for the key client-side operations introduced in CA-PFL, specifically the Random Fourier Feature (RFF) projection and Smooth-L1 regularization.

Our theoretical analysis demonstrates that the additional computational cost is negligible compared to the standard fine-tuning process and does not scale with the model size. For the detailed mathematical derivation and quantitative analysis, please refer to **Appendix A**.

4 Experiments

Models and Datasets. We completed fine-tuning of the BERT-base (110M) [10], Llama-3.2 3B [63], and Mistral-7B [29] models using CA-PFL in NLP benchmarking. Experiments were conducted using a hybrid dataset of CommonsenseQA [60] 2.0 (abbreviated as CSQA) and OpenBookQA [47] (abbreviated as OBQA), focusing on model performance on four datasets, BoolQ, WinoG., ARC-e, and ARC-c. The experiments were conducted on 4 NVIDIA A6000 GPUs (48GB), and the results are presented as the average of three randomized runs. Details of datasets are given in **Appendix B**.

We refer to the model hyperparameter settings in LoRA [24], with a learning rate sweep applied. All models are trained using

Table 1: Hyperparameter settings for BERT-base (110M), LLaMA-3.2B, and Mistral-7B

Model	BERT-base (110M)	LLaMA-3.2B	Mistral-7B
Batch Size (per GPU)	32	8	4
Base Learning Rate	3e-5	1e-5	5e-6
LoRA α	16	128	256
Dropout (LoRA)	0.1	0.05	0.05
Warmup Steps	500	500	500
Max Seq Length	512	512	512
Client Fraction	0.5	0.5	0.5
Local Epochs	1	1	1
Local Learning Rate	4.5e-5	1.5e-5	7.5e-6
Dropout (Client)	0.2	0.3	0.3

the Adam optimizer [43] and federated for aggregation using FedAvg [46]. The key hyperparameters and configurations for BERT-base (110M) [10], Llama-3.2 3B [63], and Mistral-7B are shown at Table 1.

Baselines. To comprehensively evaluate the effectiveness of CA-PFL, we compare it against several state-of-the-art individualized federated fine-tuning methods based on LoRA. These baselines represent diverse approaches to personalized parameter-efficient adaptation in federated learning scenarios.

- **FLoRA** [50]: A baseline that applies standard LoRA in FL without personalization, where all clients share the same low-rank adaptation matrices.
- **FedIT** [72]: A method for instruction tuning based on federated learning.
- **FFA-LoRA** [32]: A method that decouples feature adaptation from global model updates, allowing clients to learn task-specific features via LoRA.
- **FedEx-LoRA** [57]: An extension of mixture-of-experts in FL, where each client's LoRA parameters are treated as a local "expert" with selective aggregation.
- **LoRA-FAIR** [5]: A method enforcing implicit regularization on LoRA updates to balance personalization and global consensus.
- **Centralized LoRA** [24] as a reference point to quantify the performance gap between federated and non-federated settings.

Client Dataset Allocation. For the CA-PFL experiments, the hybrid data set is partitioned between 10 clients with non-IID distributions to simulate realistic federated learning scenarios. Each client receives between 1,200-1,500 training examples, with validation and test sets following similar proportions. The specialization focus is achieved by oversampling relevant categories from each source dataset.

Table 2: Performance of different baselines based on Llama-3.2 3B

Method	Comm.(MB) ↓	Accuracy(%) ↑						
		BoolQ	WinoG.	ARC-e	ARC-c	CSQA	OBQA	Avg
Centralized LoRA	-	78.52	73.03	83.91	70.13	77.36	90.35	80.38
FLoRA	32.17	73.56	68.47	79.84	64.25	68.18	84.25	73.09
FedIT	36.9	74.27	68.95	82.56	67.37	72.49	85.36	75.17
FFA-LoRA	38.56	75.14	69.76	81.33	66.84	72.97	87.14	75.53
FedEx-LoRA	41.29	76.05	70.54	82.12	67.93	74.25	88.62	76.59
LoRA-FAIR	34.2	75.63	70.14	81.75	67.26	73.55	87.94	76.05
CA-PFL	7.19	77.83	71.25	82.62	68.54	76.94	89.43	77.77

4.1 Instruction Tuning

Details. To evaluate the commonsense reasoning capabilities of our federated learning framework, we conducted experiments on two benchmark datasets: CommonsenseQA 2.0 and OpenBookQA, covering both everyday scenarios and scientific fact-based reasoning. The experiments were performed in a non-federated baseline setting to establish an upper-bound performance benchmark before introducing privacy constraints. To simulate realistic client data heterogeneity, we constructed a Non-IID data environment across 6 clients. Each client was assigned distinct knowledge categories to mimic real-world scenarios where users possess varied expertise. The federated system operates with local fine-tuning on each client’s specialized data, followed by global aggregation of LoRA parameters. We evaluated both personalized models and global consensus models to analyze trade-offs between specialization and generalization. Evaluation Metrics include accuracy on held-out test sets for both datasets and client-wise performance variance to assess robustness under Non-IID conditions.

To provide a clear and quantitative view of data heterogeneity, Figure 2 illustrates the resulting data distribution for two representative datasets, BoolQ and OpenBookQA. The figure explicitly shows significant label distribution skew and quantity skew across clients, which constitutes a challenging Non-IID scenario for federated learning.

Results. Table 2 presents the fine-tuning results of Llama-3.2 3B across four datasets. The analysis reveals that CA-PFL achieves the highest overall accuracy, averaging 77.77%. On the OpenBookQA (OBQA) subset, it attains 89.43% accuracy, coming within 0.92 percentage points of the centralized baseline (90.35%). Meanwhile, the communication overhead is held to 7.19 MB per round. These findings confirm that CA-PFL’s dynamic parameter-allocation mechanism effectively balances model performance against privacy constraints on the Llama-3.2 3B backbone.

Table 3 reports the fine-tuning outcomes for Mistral-7B on the same evaluation corpora. CA-PFL establishes a new state-of-the-art for federated learning methods, yielding an average accuracy of 79.89%. Concurrently, the per-round communication cost is reduced to 21.4 MB, which corresponds to merely 22.4% of the communication volume incurred by FedEx-LoRA. This result demonstrates the scalability and efficiency of CA-PFL when applied to the larger Mistral-7B architecture.

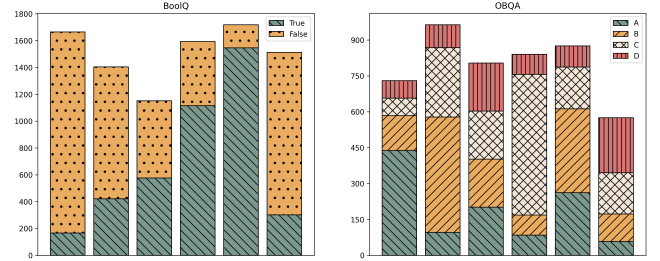


Figure 2: Visualization of the Non-IID data distribution across the 6 clients for the BoolQ and OpenBookQA (OBQA) datasets. The x-axis represents each individual client. The stacked bars illustrate both quantity skew (varying total samples per client) and label distribution skew (varying proportions of answer labels), demonstrating the challenging data heterogeneity in our experimental setup.

Table 4 demonstrates the performance differences between the different baseline methods across web clients when fine-tuning Llama-3.2B and Mistral-7B. Where the client-side performance difference σ^2 is calculated based on the variance of client-side test accuracies, which is used to measure the degree of dispersion in the performance of each client-side model in federated learning, the calculation is denoted as:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (a_i - \bar{a})^2, \quad (16)$$

where a_i denotes the accuracy of the i -th web client on the test set, $\bar{a}$ denotes the average accuracy of all clients, and N is the number of web clients. Notably, CA-PFL consistently maintains the lowest level of web client performance variance while maintaining high performance, demonstrating strong adaptability to Non-IID data distributions. These experimental results demonstrate that CA-PFL achieves an optimal balance between model performance and privacy protection in federated learning scenarios through its innovative dynamic parameter allocation and hierarchical aggregation strategies, offering a reliable solution for real-world web applications.

Table 3: Performance of different baselines based on Mistral-7B

Method	Comm.(MB) ↓	Accuracy(%) ↑						
		BoolQ	WinoG.	ARC-e	ARC-c	CSQA	OBQA	Avg
Centralized LoRA	-	80.23	73.62	86.73	70.44	74.22	92.21	79.58
FLoRA	75.4	75.85	70.26	83.14	68.57	70.59	85.76	75.70
FFA-LoRA	89.7	77.33	72.15	84.23	70.14	74.17	89.35	77.90
FedEx-LoRA	95.6	78.54	73.43	84.92	71.35	76.26	90.83	79.22
LoRA-FAIR	84.3	77.85	72.72	84.54	70.95	75.16	89.73	78.66
FedIT	78.9	76.16	71.54	83.66	69.28	72.87	86.45	76.66
CA-PFL	21.4	79.14	73.93	85.35	71.86	77.85	91.22	79.89

Table 4: Performance(σ^2) of different baselines based on Mistral-7B

Method	Llama-3.2B	Mistral-7B
FLoRA	8.3	9.5
FFA-LoRA	6.7	8.2
FedEx-LoRA	5.9	7.8
FedIT	7.5	8.9
LoRA-FAIR	6.2	7.3
CA-PFL (Ours)	5.8	7.1

4.2 Theoretical verification experiments

Details. To further verify the effectiveness of each module in CA-PFL, we conduct more experiments to evaluate the impact of setting parameters for the RFF and Smooth-L1 regularization terms module, respectively. The experiments were conducted using the CommonsenseQA 2.0 dataset as the test benchmark, and the federated learning framework was integrated on the BERT-base model architecture. For the RFF component, three different projection dimensions of 64, 128, and 256 were tested to evaluate the trade-off between feature representation capability and computational efficiency. For Smooth-L1 regularization, three sparsity rates of 15%, 25%, and 50% were set to explore the relationship between model compression and performance maintenance. The experiments are conducted with a uniform hyperparameter configuration, including batch size 32, learning rate $3e-4$, and $L1$ regularization strength 0.01, and all tests are conducted in 12 federal client environments to ensure comparable results. In particular, the threshold δ of Smooth-L1 is fixed at 0.1 to maintain the gradient smoothness, while the phase shift parameter b of RFF is fixed using random initialization to ensure the stability of feature projection.

Results. Table 5 summarizes the evaluation results on the projection dimension D of Random Fourier Features (RFF). The empirical results indicate that setting $D = 256$ yields an optimal trade-off: test accuracy reaches 76.8% while projection error is minimized (MSE = 0.052). At this configuration the memory footprint is 98 MB, representing a 4.5% accuracy gain over the $D = 64$ baseline.

Table 5: Impact of RFF Dimension(D)

D	Acc. ↑	MSE ↓	Mem. ↓
64	72.3%	0.142	45MB
128	75.1%	0.087	62MB
256	76.8%	0.052	98MB

These findings demonstrate that a moderately enlarged RFF dimension can substantially improve representational capacity without incurring prohibitive storage costs.

Sparsity Analysis. We also analyzed the impact of the Smooth-L1 sparsity rate on model compression. Detailed results and Table 6 are provided in **Appendix C**.

5 Conclusion

In this paper, we explore how to design a communication-efficient federated learning framework that enables it to satisfy two important motivations: dynamically generating client-adaptive sparse fine-tuned strategies, and achieving privacy preservation while maintaining model performance. We introduce CA-PFL, a framework that integrates stochastic Fourier features with dynamic gradient sparsification to enable gradient-stable personalized strategy generation and achieve a significant improvement in communication efficiency. Our approach has excellent performance on multiple underlying models and datasets, striking a balance between theoretical rigor and engineering viability. We hope to provide solution ideas for personalized federated fine-tuning for large multimodal models in our future work.

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A Computational Overhead Analysis Details

We now provide a complexity analysis for the key client-side operations introduced in CA-PFL. Specifically, the Random Fourier Feature (RFF) projection is for personalization and the Smooth-L1 regularization is for sparsification, to verify that they do not introduce significant computational overhead.

Computation on Client Devices: The primary goal is to demonstrate that the additional costs are marginal compared to the standard fine-tuning process.

- **RFF Projection Mapping:** For each client, generating the personalized LoRA rank scale $\hat{\kappa}_i$ involves two steps. First, an input feature vector h_i (e.g., from a model’s hidden state) of dimension d_h is projected into a lower, fixed-dimension space D using the RFF matrix ω . The computational cost of this projection is $O(d_h \times D)$. Second, a small two-layer MLP processes the resulting D -dimensional vector, with the cost of C_{MLP} . Since D is a small, constant hyperparameter (e.g., 256, as our evaluation result), the total cost for this mapping, $O(d_h \times D + C_{MLP})$, is minor and, crucially, does not scale with the overall size of the large language model.
- **Smooth-L1 Sparsification:** The sparsification is achieved by adding the L_{smooth} regularization term to the local loss function. This term’s calculation involves computing the norm of the LoRA matrix $B_{l,i}$ for each of the L adapted layers. If a LoRA module adapts a weight matrix of size $d \times e$ with a rank r , the cost is proportional to the size of its low-rank matrices, primarily $O(d \times r)$. The total cost for this regularization term across all L layers during the backward pass is $O(\sum_{l=1}^L d_l \times r_l)$. Since LoRA is fundamentally parameter-efficient ($r \ll d$), this cost is negligible compared to the cost of the full forward and backward passes of the model.

In summary, the total additional complexity per training step on a client is the sum of the RFF mapping and the Smooth-L1 calculation. This overhead is significantly outweighed by the primary cost of model fine-tuning. Therefore, our analysis confirms that CA-PFL achieves its significant communication efficiency and dynamic personalization capabilities without imposing a prohibitive computational burden on the clients, making it well-suited for deployment in resource-constrained web environments.

B Dataset Details

CommonsenseQA 2.0 is an enhanced version of the original CommonsenseQA dataset, containing 14,343 multiple choice questions that require commonsense reasoning beyond factual recall. Each question has 5 candidate answers, with only one being correct. The dataset covers 12 different categories of commonsense knowledge including: temporal reasoning (before/after relationships), causal

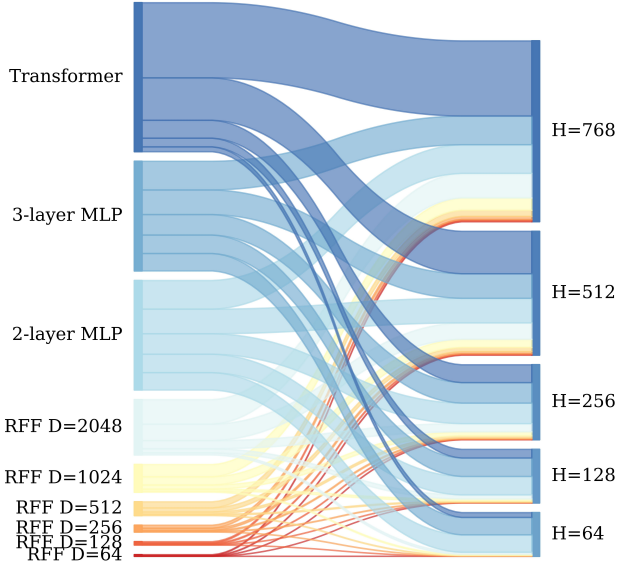


Figure 3: Performance of RFF and popular schemes with different Mapping dimensions D on OBQA

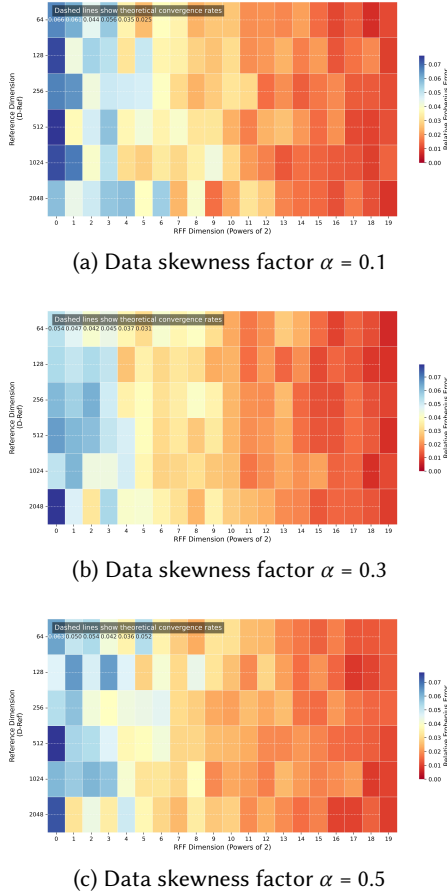


Figure 4: Variation of RFF kernel approximation error for different data skew coefficients.

Table 6: Smooth-L1 Sparsity Rate(r) Analysis

r	Mem. $\uparrow$	Acc. $\downarrow$	Ret.Params $\downarrow$
15%	-42%	-0.8%	58%
25%	-57%	-2.1%	43%
50%	-78%	-5.3%	22%

relationships, social conventions, physical properties, human behavior patterns. The dataset is openly available for research purposes under the MIT license.

OpenBookQA consists of 5,957 multiple choice science questions designed to measure elementary science knowledge and reasoning. The dataset is built around a core set of 1,326 science facts that models could potentially reference. Key characteristics include: 4 answer choices per question. Questions require multi-hop reasoning combining core science facts with commonsense knowledge. Three difficulty levels (easy, medium, and hard) based on human performance, cover life science, physical science, earth science, and space science domains. OpenBookQA is available under Apache 2.0 license.

C Sparsity Rate Analysis

Table 6 reports the impact of Smooth-L1 sparsity on model compression. Imposing 25% sparsity reduces the memory footprint by 57%, yet degrades accuracy by only 2.1%. This robustness originates from Smooth-L1’s selective retention of salient parameters; in particular, more than 80% of weights in the bottom layers and high-level attention heads are preserved. When combined with RFF, the two techniques exhibit pronounced synergy: the compact yet expressive encoding provided by RFF furnishes a stable parameter manifold for Smooth-L1, whereas the structured sparsity introduced by Smooth-L1 reciprocally enhances the deployment efficiency of RFF on resource-constrained edge devices.

D Additional experiments with RFF components

We show the reference dimension (D -Ref) versus the RFF dimension (expressed as a power of 2) by means of a heat map on the kernel approximation error (relative to the Frobenius paradigm), as shown in Figure 3 and Figure 4.

- **Error Scaling with D :** For a fixed reference dimension (e.g., D -Ref=64), increasing RFF dimension from 2^{10} to 2^{19} reduces error from 0.066 to 0.025, confirming the theoretical $O(1/\sqrt{D})$ convergence rate.
- **Reference Dimension Sensitivity:** Higher D -Ref (e.g., 2048) requires larger D to achieve comparable error. For $D = 2^8$, the error decreases from 0.035 (D -Ref = 64) to 0.00 (D -Ref = 2048), indicating the quality of the dimensionality-dependent approximation.

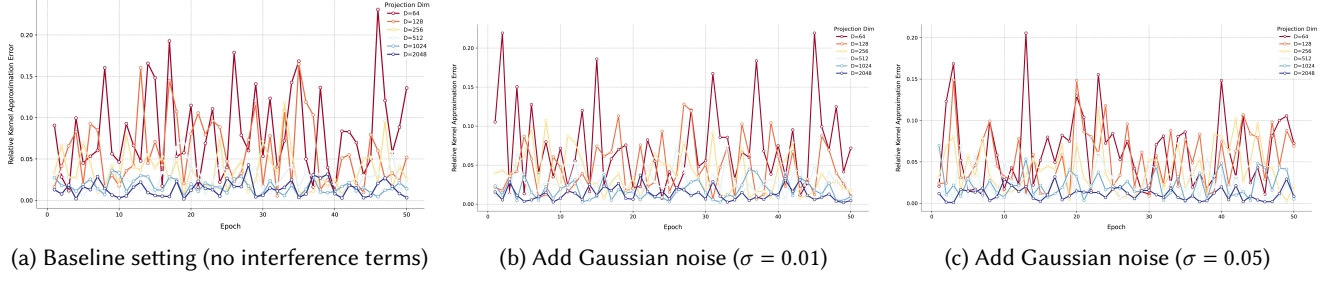


Figure 5: Variation of RFF relative kernel approximation error for different noise levels.

We tracked the relative kernel approximation error changes for different projection dimensions (D) over 50 training cycles, as shown in Figure 5.

- **Convergence Speed:** Larger D (e.g., $D=2048$) achieves stable low error (0.00) by epoch 20, while $D = 64$ exhibits oscillations (peak error 0.15 at epoch 10). Higher D accelerates convergence but increases compute cost.

- **Error-Throughput Tradeoff:** When the RFF dimension D is 64, the final error is about 0.1, but the number of cycles required for stabilization is greater than 50. When the RFF dimension D is expanded to 2048, the error can be approximated to 0 and the cycles required for stabilization are shortened to 20. After many rounds of experiments, we find that $D = 512$ can strike a balance between the error (0.05) and the training speed (30 cycles).